Sensor-Fusion System

Iteration 2 - System Design and Implementation
Updated from the October 13, 2026 Iteration 1 project plan

Project	Team / course
Sensor-Fusion Simulation (C++17)	Diego Juarez (Project Leader) • Anthony Santana
Repository	github.com/Juarez-Diego-2007/sensor-fusion-simulation
Integration status	feature/anthony-evaluation reviewed by Diego and merged into main
Updated status	C++ program tested; Diego completed review and merged changes to main. Documentation upload and individual Canvas submissions remain to verify.

1. Project scope and expected deliverables

This iteration turns the Iteration 1 plan into a functioning, text-based system. The program simulates two sensors measuring the distance to an obstacle, uses a weighted average to combine readings, and compares the fused estimate against the known ground-truth distance. Physical sensors, external graphics, and advanced filtering remain outside scope.

Core acceptance criteria: a C++17 program that compiles and runs; all seven tasks present in the switch-based menu; separate classes and files; a class/system diagram; a complete task register; evidence of each team member's contributions; and a repository containing README and docs/ with PDF and Excel. The tested C++ work has been reviewed and merged; repository documentation and individual submissions must be checked separately.

2. Iteration leadership and responsibilities

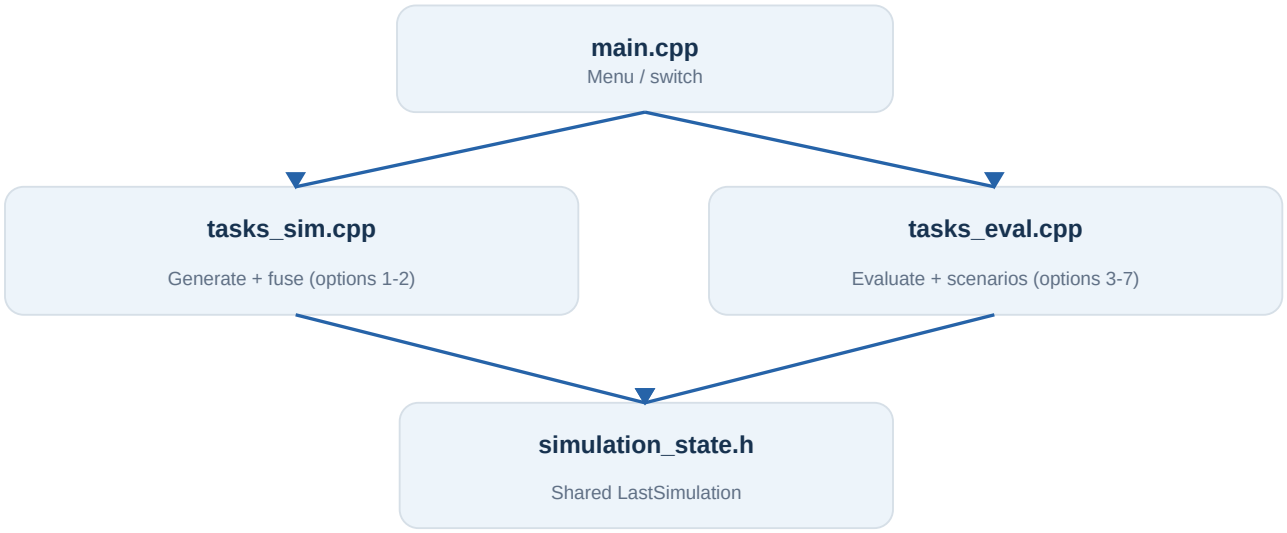
Team member	Role and verified contribution
Diego Juarez	Iteration 2 Project Leader. Wrote the initial Sensor class, FusionEngine, and two simulation task functions. Reviewed the integration pull request and merged the completed code into main.
Anthony Santana	Leads quantitative evaluation and integration. Built the Evaluation class, error comparisons, three scenarios, switch-based menu, saved-results display, shared-data integration, and manual test runs.
Both	Reviewed and integrated the code. Finalize the task tracker and report, confirm the README and docs/ files are on main, and submit individual repository links on Canvas.

Project leadership status

Diego is the designated Project Leader for Iteration 2. The team confirmed that he reviewed the code and merged the pull request into the main branch. A check-in presentation and individual Canvas submissions are not documented as completed.

3. System architecture and data flow

A single shared result holds the latest sample so that generation, fusion, and evaluation compare the same pair of readings. The menu dispatches exactly one task function for each user choice.



Class relationships: Sensor generates noisy readings; FusionEngine combines them; Evaluation compares the same readings and fused value against ground truth. LastSimulation stores the most recent values for the task functions.

4. Classes, private data, and public methods

Class / type	Private data	Public interface
Sensor	name; noiseLevel; weight; mt19937 generator	Sensor(name, noise, weight); generateReading(actualDistance); getWeight(); getName()
FusionEngine	None (utility class)	static weightedAverage(reading1, weight1, reading2, weight2)
Evaluation	sensor1Error; sensor2Error; fusedError	Evaluation(actualDistance, reading1, reading2, fusedEstimate); static absoluteError(actual, measured); printReport()
LastSimulation (struct)	None; stored fields are public	available; scenario; actualDistance; reading1; reading2; fusedEstimate; weight1; weight2; fusedAvailable

Data structure choice: **LastSimulation** is a struct rather than a class. It has public fields and is defined in **simulation_state.h**; exactly one shared variable is defined in **tasks_eval.cpp** and accessed using an extern declaration.

5. Complete menu and task register

Status reflects the completed code tests and the team-confirmed Diego review / merge. Menu number 0 exits the program; non-menu deliverables are listed separately.

ID / option	Task and responsible lead	Files / function	Deliverable / status
T01 / 1	Generate sensor readings - Diego	tasks_sim.cpp; generateSensorReadings()	Saved reading pair; Done
T02 / 2	Calculate fused estimate - Diego	tasks_sim.cpp; calculateFusedEstimate()	Fusion on saved sample; Done
T03 / 3	Calculate errors - Anthony	evaluator.h/.cpp; tasks_eval.cpp; calculateEstimationErrors()	Absolute errors and comparisons; Done
T04 / 4	Low-noise case - Anthony	tasks_eval.cpp; runLowNoiseScenario()	Scenario + error report; Done
T05 / 5	High-noise case - Anthony	tasks_eval.cpp; runHighNoiseScenario()	Scenario + error report; Done
T06 / 6	Outlier case - Anthony	tasks_eval.cpp; runOutlierScenario()	Fault scenario + report; Done
T07 / 7	Display saved results - Anthony	tasks_eval.cpp; displaySimulationResults()	Stored sample + errors; Done
T08 / 0	Menu / exit - Anthony	main.cpp; main()	Switch dispatch + exit; Done
T09 / -	Shared-data integration - Anthony; Diego reviewed	simulation_state.h; tasks_sim.cpp; tasks_eval.cpp	Consistent saved readings; reviewed and merged to main; Done

Additional required work

ID	Non-menu deliverable	Responsibility	Status
T10	Class diagram and system design	Both	Prepared in this report; document review not yet confirmed
T11	Compile + manual scenario testing	Anthony; Diego reviewed integration	Manual testing done; code reviewed
T12	Updated README	Anthony lead	Prepared; GitHub upload pending
T13	Revised Excel plan	Both	Prepared; GitHub upload pending
T14	Iteration 2 PDF report	Both	Prepared; GitHub upload pending
T15	PR review and merge	Diego (leader), Anthony	Done - reviewed by Diego and merged into main
T16	docs/ upload and Canvas links	Both / each student	Upload and individual submission not confirmed

6. File structure and function ownership

File	Purpose / lead
src/main.cpp	Input validation, while loop, switch menu, exit - Anthony
src/tasks.h	Declarations for all seven menu functions - Diego drafted
src/tasks_sim.cpp	Generate readings and calculate saved fusion - Diego initial; Anthony integrated
src/tasks_eval.cpp	Errors, three scenarios, saved results, single shared LastSimulation definition - Anthony
src/simulation_state.h	Shared result structure and extern state declaration - Anthony integration
src/sensor.h / sensor.cpp	Sensor private attributes, random reading generator - Diego
src/fusion.h / fusion.cpp	Weighted average and weight validation - Diego
src/evaluator.h / evaluator.cpp	Error calculations and comparison reporting - Anthony
README.md	Description, dependencies, command-line build/run, menu, test instructions - Anthony lead
docs/*.pdf / docs/*.xlsx	System-design PDF and revised Excel plan - both members

7. Implementation and menu behavior

• **Option 1:** create a new sample at 100 cm using noise levels 2.0 and 5.0; save sensor readings and weights 0.7/0.3; mark fusion not yet calculated.

• **Option 2:** require a saved sample, then calculate and save one weighted-average estimate from those exact readings.

• **Option 3:** require a saved fused estimate; calculate absolute error for Sensor 1, Sensor 2, and fusion from the stored ground truth.

• **Options 4-6:** generate low-noise (0.5/1.0), high-noise (8.0/15.0), or outlier (standard noise and +40 cm to Sensor 2) samples; fuse, save, and evaluate each case.

• **Option 7:** redisplay the most recent saved readings; show a reminder when a fused estimate is not yet available.

Algorithm: fused estimate = (w1 x r1 + w2 x r2) / (w1 + w2). Absolute error = |ground truth - estimate|. Weights are assigned constants, not learned. A lower absolute error indicates higher accuracy for a particular sample.

Build command (from repository root)

```
g++ -std=c++17 -Wall -Wextra -pedantic src/main.cpp src/sensor.cpp src/fusion.cpp src/tasks_sim.cpp
src/evaluator.cpp src/tasks_eval.cpp -o sensor_fusion
Run: ./sensor_fusion
```

8. Manual testing and evidence

The program compiled successfully with C++17 and the team recorded working menu, invalid-input handling, and saved-sample behavior. Values below are observed example runs from Anthony's terminal screenshots; random noise means another run will differ.

Scenario	Truth	Sensor 1	Sensor 2	Fused	Errors: S1 / S2 / Fusion
Standard	100.00	96.88	100.07	97.84	3.12 / 0.07 / 2.16
Low noise	100.00	100.07	99.79	99.99	0.07 / 0.21 / 0.01
High noise	100.00	91.69	122.86	101.04	8.31 / 22.86 / 1.04
Outlier	100.00	99.77	142.64	112.63	0.23 / 42.64 / 12.63

Verification	Observed outcome
7 before data	Reported no saved simulation available
1 -> 7	Displayed the same saved readings and indicated fusion was not yet calculated
1 -> 2 -> 3 -> 7	Readings matched across menu tasks; fused value 97.84 cm and errors were consistent
4, 5, 6 -> 7	Option 7 redisplayed the corresponding scenario without generating new readings
Invalid choice / exit	Invalid numeric option handled; option 0 exited normally
Version control	Anthony pushed the integration branch; Diego reviewed and merged the PR into main (team-confirmed)

9. Interpretation, limitations, and risks

- The low-noise and high-noise representative runs showed lower fused error than either individual sensor, but this is not guaranteed for every random sample.
- The outlier case degraded fusion relative to the healthy sensor: fused error 12.63 cm versus Sensor 1 error 0.23 cm. This demonstrates the sensitivity of a fixed weighted average to faulty readings.
- No fixed random seed or automated multi-run statistical evaluation was documented. Manual examples demonstrate functionality but not population-level performance.
- Other limitations: only two sensors, simulated distance, fixed weights, no real device input, no advanced outlier rejection.

10. Revised timeline and milestones

The original Iteration 1 plan contained proposed dates. These are retained where applicable and updated with the work evidenced by the Iteration 2 code/test record. November 2 remains a proposed final review date, not an independently verified Canvas deadline.

Dates	Milestone / owner	Deliverable / latest status
Oct 13-15	Define initial scope - both	Iteration 1 plan prepared October 13
Oct 16-22 (planned)	Sensor + fusion implementation - Diego	Sensor, FusionEngine and simulation tasks; reported complete
Oct 23-30	Errors, scenarios and menu - Anthony	All seven menu tasks implemented and manually tested
After implementation	Shared reading integration and review - Anthony, Diego	Same sample across options 1, 2, 3, 7; Diego reviewed and merged PR to main
Oct 30-Nov 1 (planned)	README, diagram, Excel and PDF - both	Files prepared; docs/ commit still to be confirmed
Nov 2 (proposed)	Final documents and submission - both	PR merged into main; docs/ check and Canvas links not yet confirmed

11. Communication, task tracking, and challenge management

Communication and progress tracking follow the Iteration 1 approach: direct team messaging for daily coordination, GitHub branches and pull requests for code changes, and the revised Excel task register for task ownership, status, milestone dates, and evidence. As Project Leader, Diego coordinates daily progress, balances tasks, presents the check-in status, and maintains the task tracker. The team should record any actual meeting times and decisions rather than inventing them.

Integration and review completed: originally, generating readings and calculating a fused estimate each created different random samples. The shared LastSimulation state now reuses one pair of readings. Diego has reviewed the implementation changes and merged the pull request into main. **Remaining closeout:** verify documentation files on main and complete individual Canvas submissions.

12. Contribution record and final repository checklist

Member	Completed work / evidence	Outstanding
Diego Juarez	Initial sensor/noise modeling, weighted fusion, tasks 1-2. Iteration 2 Project Leader; reviewed code and merged pull request into main.	Confirm documentation and check-in evidence, if required.
Anthony Santana	Evaluator, menu, three scenarios, saved output, shared-data integration; compiled/tested, committed and pushed branch.	Verify docs/ and README on main; submit repository link.
Both	Original scope and design decisions; implementation integrated and merged on main after Diego review.	Confirm task statuses and dates; finalize docs and README; individually submit links.

Repository requirements: C++ source files; updated README with build/run instructions; docs/Sensor_Fusion_Iteration2_System_Design_Report.pdf; docs/Sensor_Fusion_Iteration2_Project_Plan.xlsx; merged main branch (completed); and an individual Canvas submission by each student (not yet confirmed).